

Karmesh Yadav

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Education

Georgia Institute of Technology

PHD IN COMPUTER SCIENCE

- Advisors: Dr. Zsolt Kira and Dr. Dhruv Batra

Atlanta

Aug 2023 - Aug 2026 (Expected)

Carnegie Mellon University

MASTERS IN ROBOTIC SYSTEMS DEVELOPMENT

- Capstone Project Advisor: Dr. John Dolan

Pittsburgh

Aug 2018 - May 2020

Indian Institute of Technology, Guwahati

B.TECH. IN MECHANICAL ENGINEERING

- Undergrad research advisor: Dr. S.K. Dwivedi

Guwahati

July 2013 - May 2017

Professional Experience

Mistral AI

AI SCIENTIST INTERN

- Initiated and led the Multimodal Reasoning effort, establishing end-to-end training, evaluation, and data pipelines.
- Developing rubric-guided reinforcement learning methods to reduce visual hallucinations and improve grounded reasoning in multimodal language models.

Palo Alto

May 2025 - Current

Yutori

INTERN, TECHNICAL STAFF, AI

- Developed deployment pipelines for web agents to operate on real websites while effectively avoiding bot detection.
- Created data filtering pipeline and trained VLM-based agents for web navigation tasks.

San Francisco

Aug 2024 - Nov 2024

Fundamental AI Research (FAIR), Meta AI

AI RESIDENT

- Researched self-supervised pretraining techniques for learning useful visual representations for embodied agents.
- Released the HM3D-Semantics (HM3DSem) dataset and the Open-Vocabulary Mobile Manipulation (OVMM) benchmark based on the Habitat Simulator.

Menlo Park

Aug 2021 - Jun 2023

ISEE Inc.

SENIOR ROBOTICS ENGINEER

- Explored deep uncertainty estimation techniques for predicting the closed loop tracking performance of an AV controller. Estimated the collision prob. of the AV w.r.t. obstacles in an occupancy grid.
- Improved the trajectory optimization planner and robustified its collision checking. This led to an increased confidence in its performance and resulted in its deployment on the AV.

Boston

Jul 2020 - Aug 2021

ISEE Inc.

SOFTWARE DEVELOPMENT INTERN

- Built toolboxes to automate the system identification and calibration procedure of Isee's vehicles. Also researched and implemented various vehicle and tire models for control application in AVs.

Boston

May 2019 - Aug 2019

Autonomous Driving Team, Mathworks

INTERN

- Worked on improving the localization module of the autonomous vehicle by fusing ORB-SLAM's output with other sensors like GPS, IMU and Wheel Odometer

Hyderabad

Aug 2017 - Nov 2017

Publications

CONFERENCE PUBLICATION

- Karmesh Yadav***, Yusuf Ali*, Yarin Gal, Zsolt Kira. FindingDory: A Benchmark to Evaluate Memory in Embodied Agents. arXiv preprint arXiv:2506.15635, (ECCV) 2026.
- Moises Andrade, Joonhyuk Cha, Brandon Ho, Vriksha Srihari, **Karmesh Yadav**, Zsolt Kira. Let's Think in Two Steps: Mitigating Agreement Bias in MLLMs with Self-Grounded Verification. International Conference on Learning Representations (ICLR) 2026.
- Gunshi Gupta, **Karmesh Yadav**, Zsolt Kira, Yarin Gal, Rahaf Aljundi. Memo: Training Memory-Efficient Embodied Agents with Reinforcement Learning. **Spotlight (Selection Rate: 688/21575 = 3.2%)** at Neural Information Processing Systems (NeurIPS), 2025.
- Gunshi Gupta*, **Karmesh Yadav***, Yarin Gal, Dhruv Batra, Zsolt Kira, Cong Lu, Tim G. J. Rudner. Pre-trained Text-to-Image Diffusion Models Are Versatile Representation Learners for Control. **Spotlight (Selection Rate: 326/15671 = 2.1%)** at Neural Information Processing Systems (NeurIPS), 2024 and **Oral** at Workshop on Generative Models for Decision Making (GenAI4DM) at ICLR 2024.
- Arjun Majumdar*, Anurag Ajay*, Xiaohan Zhang*, Sriram Yenamandra, Mikael Henaff, Alexander Sax, Sneha Silwal, Paul Mcvay, Oleksandr Maksymets, Sergio Arnaud, Pranav Putta, **Karmesh Yadav**, Qiyang Li, Ben Newman, Mohit Sharma, Vincent-Pierre Berges, Shiqi Zhang, Pulkit Agrawal, Dhruv Batra, Yonatan Bisk, Mrinal Kalakrishnan, Franziska Meier, Chris Paxton, Aravind Rajeswaran. OpenEQA: Embodied Question Answering in the Era of Foundation Models. Conference on Computer Vision and Pattern Recognition (CVPR), 2024 and **Spotlight** at ICRA Workshop on Mobile Manipulation and Embodied Intelligence, 2024
- Sneha Silwal*, **Karmesh Yadav***, Tingfan Wu*, Jay Vakil*, Arjun Majumdar*, Sergio Arnaud*, Claire Chen, Vincent-Pierre Berges, Dhruv Batra, Aravind Rajeswaran, Mrinal Kalakrishnan, Franziska Meier, Oleksandr Maksymets. What do we learn from a large-scale study of pre-trained visual representations in sim and real environments? International Conference on Robotics and Automation (ICRA), 2024 and **Spotlight** at ICRA Workshop on Mobile Manipulation and Embodied Intelligence, 2024
- Sriram Yenamandra*, Arun Ramachandran*, **Karmesh Yadav***, Austin Wang, Mukul Khanna, Theophile Gervet, Tsung-Yen Yang, Vidhi Jain, Alexander William Clegg, John Turner, Zsolt Kira, Manolis Savva, Angel Chang, Devendra Singh Chaplot, Dhruv Batra, Roozbeh Mottaghi, Yonatan Bisk, Chris Paxton. HomeRobot: Open-Vocabulary Mobile Manipulation. Conference on Robot Learning (CoRL), 2023.
- Arjun Majumdar*, **Karmesh Yadav***, Sergio Arnaud*, Yecheng Jason Ma, Claire Chen, Sneha Silwal, Aryan Jain, Vincent-Pierre Berges, Pieter Abbeel, Jitendra Malik, Dhruv Batra, Yixin Lin, Oleksandr Maksymets, Aravind Rajeswaran, Franziska Meier. Where are we in the search for an Artificial Visual Cortex for Embodied Intelligence? Neural Information Processing Systems (NeurIPS), 2023.
- Jacob Krantz, Theophile Gervet, **Karmesh Yadav**, Austin Wang, Chris Paxton, Roozbeh Mottaghi, Dhruv Batra, Jitendra Malik, Stefan Lee, Devendra Singh Chaplot. Navigating to Objects Specified by Images. International Conference on Computer Vision (ICCV), 2023.
- Karmesh Yadav***, Ram Ramrakhya*, Santhosh Kumar Ramakrishnan*, Theo Gervet, John Turner, Aaron Gokaslan, Noah Maestre, Angel Xuan Chang, Dhruv Batra, Manolis Savva, Alexander William Clegg, Devendra Singh Chaplot. Habitat-Matterport 3D Semantics Dataset. Conference on Computer Vision and Pattern Recognition (CVPR), 2023. **Highlight Paper (Selection Rate: 235/9155 = 2.5%)**
- Justin Wasserman, **Karmesh Yadav**, Girish Chowdhary, Abhinav Gupta, Unnat Jain. Last-Mile Embodied Visual Navigation. Conference on Robot Learning (CoRL), 2022.
- Gunshi Gupta*, **Karmesh Yadav***, Liam Paull. Look-ahead meta learning for continual learning. Neural Information Processing Systems (NeurIPS), 2020. **Oral (Selection Rate: 105/9454 = 1.1%)**
- Vignesh Prasad*, **Karmesh Yadav***, Rohitashva Singh Saurabh, Swapnil Daga, Nahas Pareekutty, K Madhava Krishna, Balaraman Ravindran, Brojeshwar Bhowmick. Learning to Prevent Monocular SLAM Failure using Reinforcement Learning. Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP), 2018.

WORKSHOP PUBLICATION

- Karmesh Yadav**, Ram Ramrakhya, Arjun Majumdar, Vincent-Pierre Berges, Sachit Kuhar, Dhruv Batra, Alexei Baevski, and Oleksandr Maksymets. Offline visual representation learning for embodied navigation. Reincarnating Reinforcement Learning (RRL) at ICLR, 2023.

PREPRINT

Alexander H. Liu, Andy Ehrenberg, Andy Lo, Chen-Yo Sun, Guillaume Lample, Jean-Malo Delignon, ..., **Karmesh Yadav**, Voxtral Realtime. arXiv preprint arXiv:2602.11298, 2026.

Alexander H. Liu, Kartik Khandelwal, Sandeep Subramanian, Victor Jouault, Abhinav Rastogi, ..., **Karmesh Yadav**, Minis-tral 3. arXiv preprint arXiv:2601.08584, 2026.

Alexander H. Liu, Andy Ehrenberg, Andy Lo, Clément Denoix, Corentin Barreau, Guillaume Lample, ..., **Karmesh Yadav**, Voxtral. arXiv preprint arXiv:2507.13264, 2025.

Abhinav Rastogi, Albert Q. Jiang, Andy Lo, Gabrielle Berrada, Guillaume Lample, ..., **Karmesh Yadav**, Magistral. arXiv preprint arXiv:2506.10910, 2025.

Ahmad Elawady, Gunjan Chhablani, Ram Ramrakhya, **Karmesh Yadav**, Dhruv Batra, Zsolt Kira, Andrew Zsolt. ReLIC: A recipe for 64k steps In-Context Reinforcement Learning for Embodied AI. arXiv preprint arXiv:2410.02751, 2024.

Karmesh Yadav*, Arjun Majumdar*, Ram Ramrakhya, Naoki Yokoyama, Alexei Baevski, Zsolt Kira, Oleksandr Maksymets, Dhruv Batra. OVRL-V2: A simple state-of-art baseline for ImageNav and ObjectNav. arXiv preprint arXiv:2303.07798, 2023.

Service & Teaching

WORKSHOPS AND COMPETITIONS

May 2024	Workshop on Vision-Language Models for Mobile Manipulation , Organizer	<i>Yokohama</i>
Dec 2023	Open Vocabulary Mobile Manipulation Challenge, NeurIPS , Organizer	<i>New Orleans</i>
Jun 2023	Habitat Navigation Challenge, CVPR , Lead Organizer	<i>Vancouver</i>
Dec 2022	Habitat Rearrangement Challenge, NeurIPS , Organizer	<i>New Orleans</i>
Jun 2022	Habitat ObjectNav Challenge, NeurIPS , Lead Organizer	<i>New Orleans</i>

PEER REVIEW

NeurIPS'23, ICLR'24, ICLR'25, CVPR'25, ICLR'26, ECCV'26

MENTORING

2026	Shivang Chopra , PhD Student	<i>Georgia Tech</i>
2024-2026	Yusuf Ali , PhD Student	<i>Georgia Tech</i>
2024-2025	Moises Andrade , PhD Student	<i>Georgia Tech</i>
2022-2023	Sriram Yenamandra , Masters Student	<i>Georgia Tech</i>
2022-2023	Arun Ramachandran , Masters Student	<i>Georgia Tech</i>

TEACHING

Fall'25	Deep Learning , Teaching Assistant	<i>Georgia Tech</i>
Spring'24	Deep Learning , Teaching Assistant	<i>Georgia Tech</i>